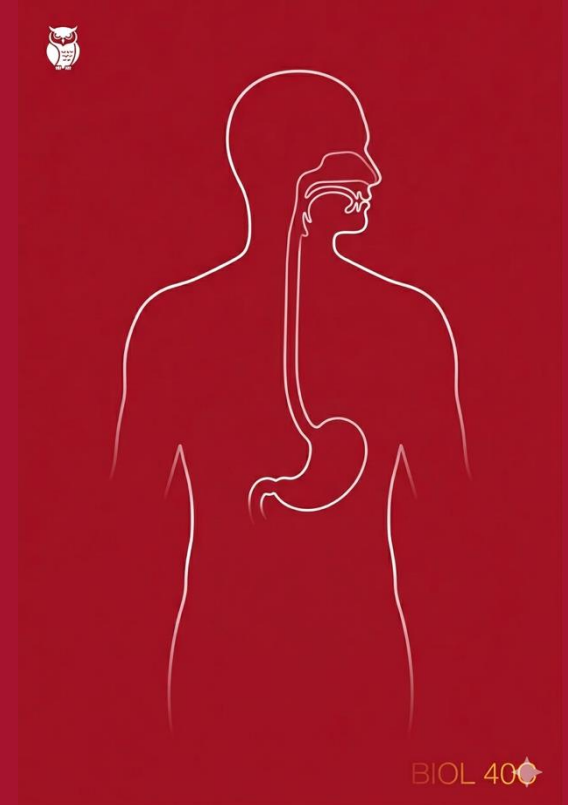

BIOL 40C

Human Anatomy & Physiology III

Class 2: The Upper GI Tract — Mouth to Stomach

Spring 2026 | Foothill College

Giorgio Lagna, Ph.D. | Wednesday, April 8, 2026



Today's Journey

BIOL 40C

1

Quick Review + Oral Cavity

10:00–10:25

2

Esophagus + Transport Review

10:25–10:55

3

Gastric Acid Secretion

11:05–11:35

4

GERD: Clinical Application

11:35–11:50

ACTIVITY

Turn to a neighbor and name all four layers of the alimentary canal, from innermost to outermost.

No notes allowed! Work together. You have 2 minutes.

Hint: Think from the inside of the tube outward.

The Four Layers

BIOL 40C

1

Mucosa

Epithelium + lamina propria + muscularis mucosae — faces the lumen

2

Submucosa

Dense CT, blood/lymph vessels, submucosal plexus — the supply layer

3

Muscularis Externa

Inner circular + outer longitudinal smooth muscle — the movement layer

4

Serosa / Adventitia

Outer covering — serosa (intraperitoneal) or adventitia (retroperitoneal)

The Oral Cavity: Where It All Begins

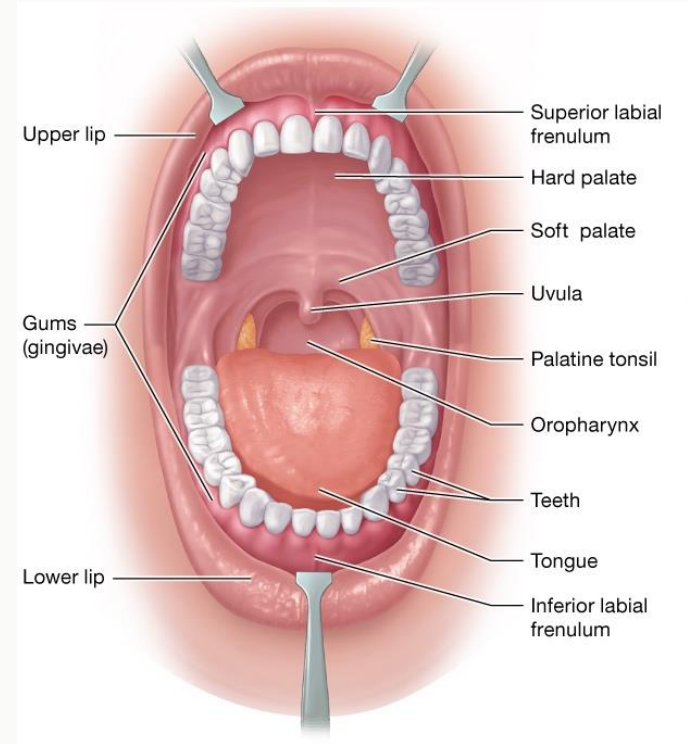
BIOL 40C

CONCEPT

Key Structures

- Hard palate and soft palate
- Uvula
- Tongue (manipulates food, taste receptors)
- Teeth (mechanical digestion begins here)

The mouth is the entry point for both mechanical and chemical digestion.



Teeth: Mechanical Digestion Starts Here

BIOL 40C

CONCEPT

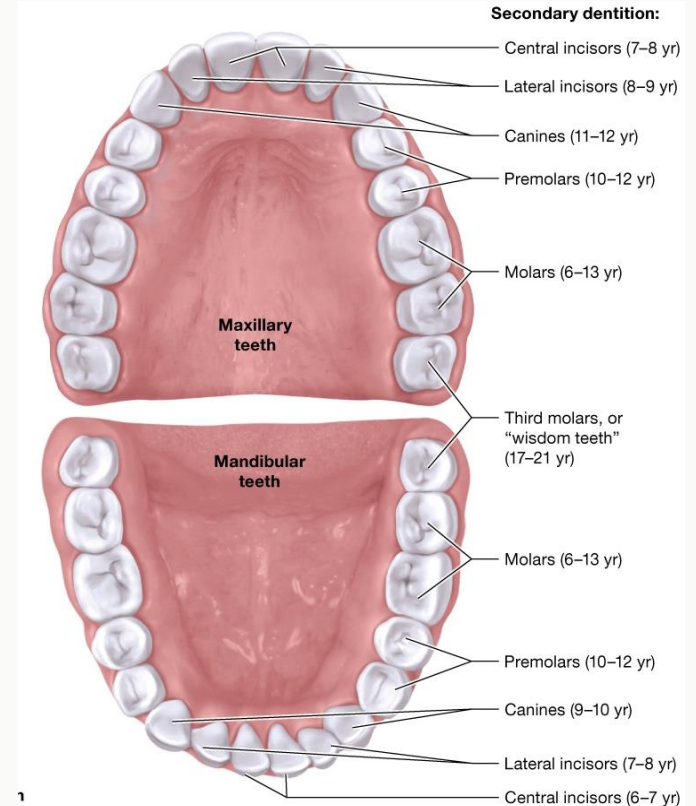
Tooth Types and Functions

Incisors Cutting

Canines Tearing

Premolars Crushing

Molars Grinding



ACTIVITY

You bite into an apple.

What's happening mechanically?
What's happening chemically?

Discuss with your neighbor. You have 2 minutes.

Mechanical

- Incisors cut through skin
- Molars grind the flesh
- Tongue mixes food with saliva
- Bolus formation

Chemical

- Salivary amylase breaks down starch
- Lingual lipase begins fat digestion
- Lysozyme kills bacteria
- Water dissolves molecules for taste

Salivary Glands

BIOL 40C

CONCEPT

Parotid

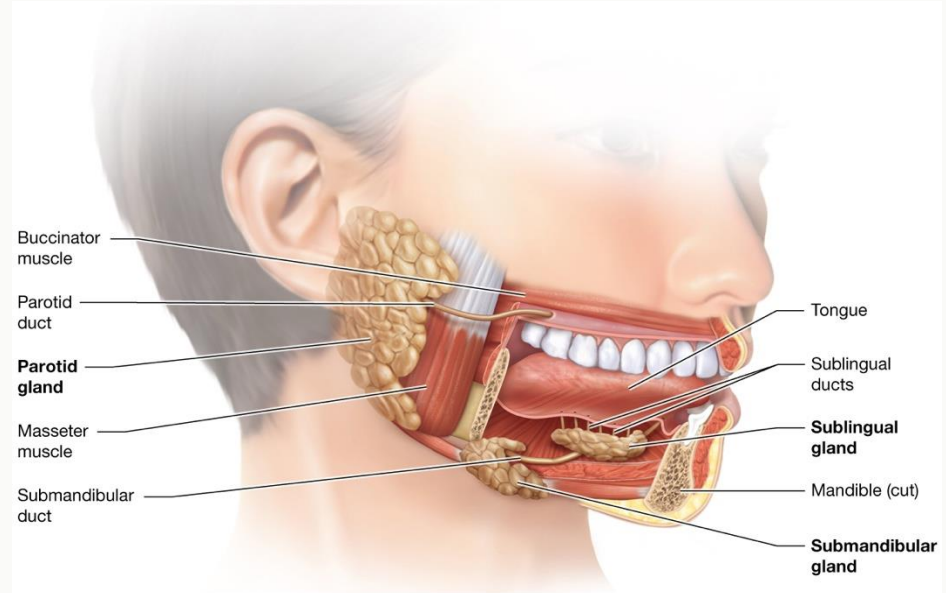
Largest; anterior to ear; serous (watery) secretion rich in amylase

Submandibular

Below mandible; mixed serous and mucous secretion

Sublingual

Under tongue; mostly mucous secretion



Saliva: More Than Just Water

BIOL 40C

CONCEPT

Saliva has multiple roles: digestion, protection, and lubrication.

Water (99.5%)

Dissolves food molecules for taste; moistens bolus

Salivary amylase

Begins starch digestion (breaks starch into maltose)

Lingual lipase

Begins fat digestion (activated in stomach)

Mucus

Lubricates food for swallowing; protects oral mucosa

Lysozyme + IgA

Antimicrobial defense (innate + adaptive immunity)

Swallowing (Deglutition): Three Phases

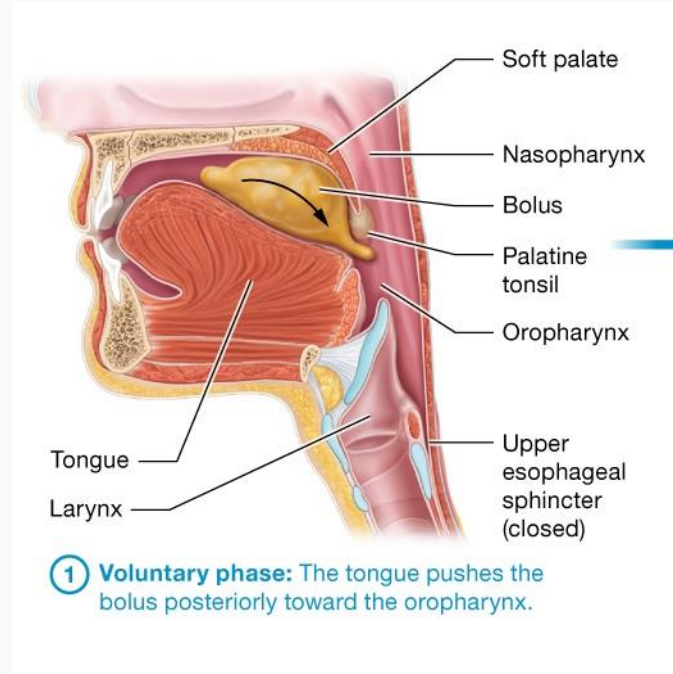
BIOL 40C

CONCEPT

1

Voluntary Phase

Tongue pushes bolus posteriorly toward oropharynx. You decide to swallow.



Swallowing (Deglutition): Three Phases

BIOL 40C

CONCEPT

Voluntary Phase

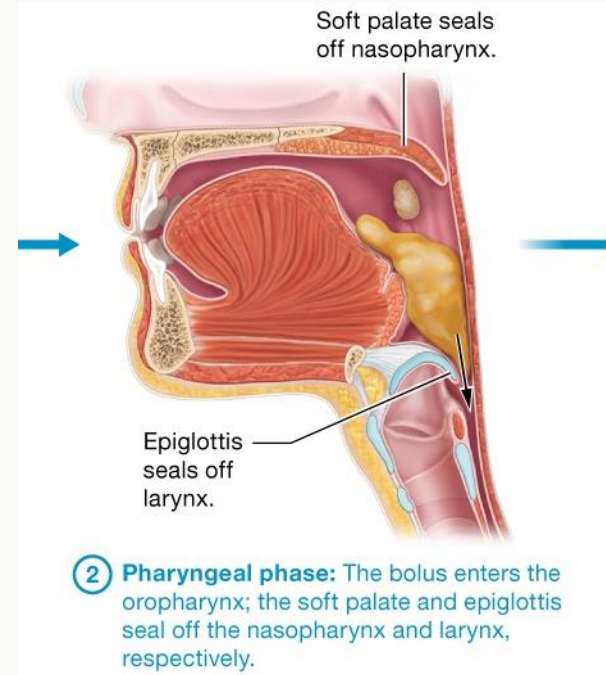
1

Tongue pushes bolus posteriorly toward oropharynx. You decide to swallow.

Pharyngeal Phase

2

Reflexive. Soft palate seals nasopharynx. Epiglottis covers larynx. Bolus enters esophagus.



Swallowing (Deglutition): Three Phases

BIOL 40C

CONCEPT

Voluntary Phase

1

Tongue pushes bolus posteriorly toward oropharynx. You decide to swallow.

Pharyngeal Phase

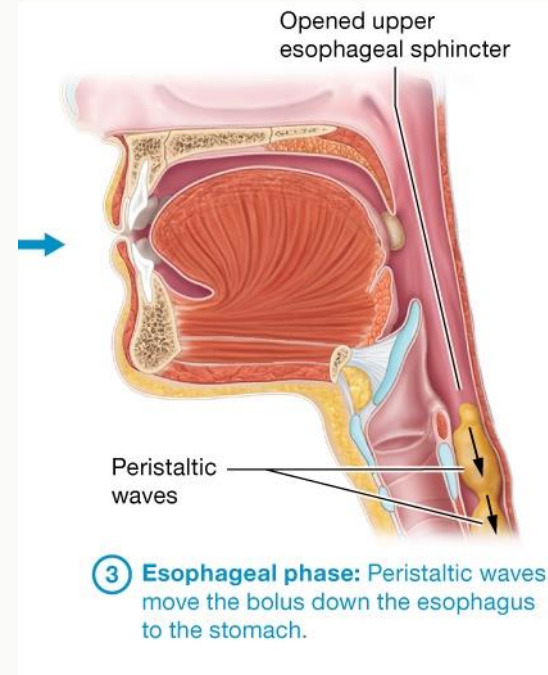
2

Reflexive. Soft palate seals nasopharynx. Epiglottis covers larynx. Bolus enters esophagus.

Esophageal Phase

3

Peristalsis takes over. Bolus travels to stomach in 4–8 seconds. Entirely involuntary.



ACTIVITY

Try this right now:

Swallow once.

Now try to swallow again immediately.

Why is the second swallow harder?

No bolus to trigger the pharyngeal reflex. The reflex needs sensory input from food touching the back of the pharynx.

Esophagus: Structure and Function

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CONCEPT

~25 cm muscular tube from pharynx to stomach

Mucosa

Stratified squamous epithelium (protection from abrasion)

Muscularis

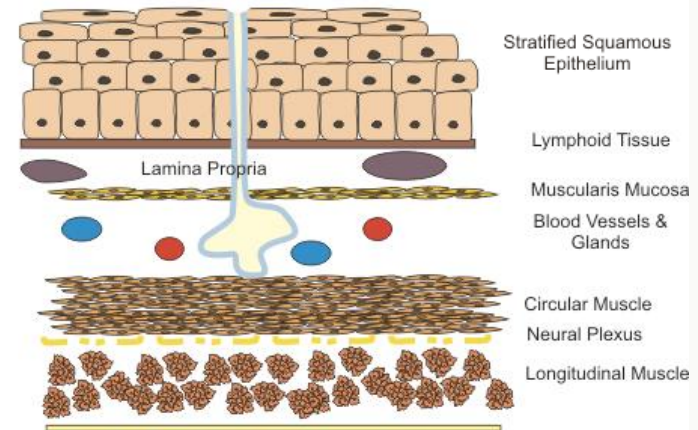
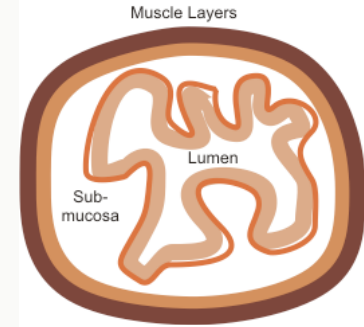
Upper 1/3: skeletal muscle (voluntary)
Lower 2/3: smooth muscle (involuntary)

Function

Peristalsis transports bolus to stomach

Key Question

What is it NOT well-protected against?
(Chemical damage = acid)



Peristalsis: Moving Food Along

BIOL 40C

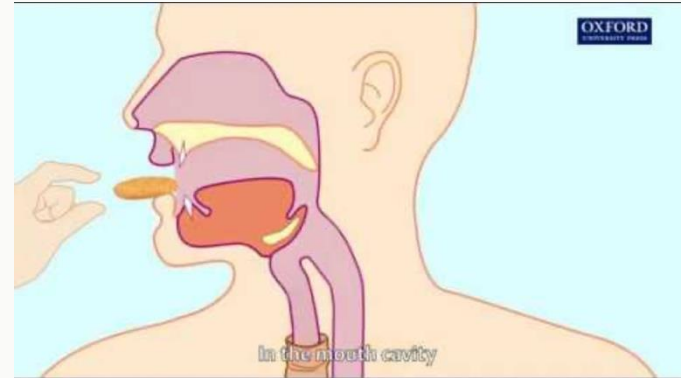
CONCEPT

How It Works

Circular muscles contract behind the bolus (squeezing).
Longitudinal muscles contract ahead (shortening the tube).
The result: a wave that propels food forward.

Analogy

Like squeezing a tube of toothpaste from the bottom.
The bolus can only go one direction.



The Lower Esophageal Sphincter (LES)

BIOL 40C

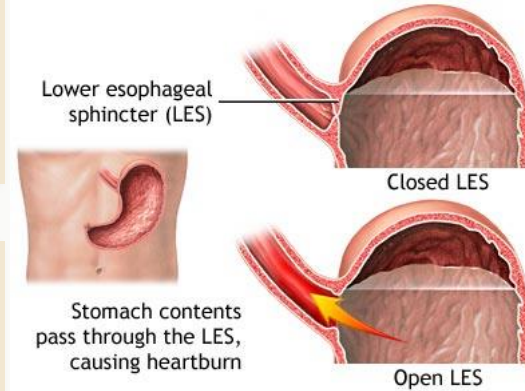
CONCEPT

Gatekeeper between esophagus and stomach

Normally closed to prevent backflow of acidic stomach contents.
Opens when the peristaltic wave arrives with food.

When the LES fails:

Acid reflux (GERD). The stratified squamous epithelium of the esophagus is not built to handle acid.
We'll return to this at the end of class with a clinical case.

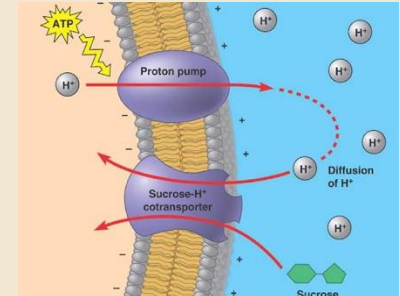
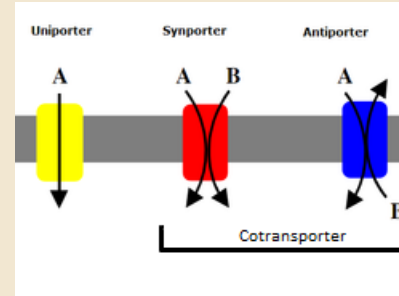


A&P REVIEW

Before we dive into the stomach, we need to refresh some transport concepts from 40A.

These are going to be **EVERYWHERE** this quarter.

- The stomach uses an H^+/K^+ ATPase to **pump** acid.
- The intestines use **co-transporters** to absorb glucose and amino acids.
- Every nutrient that enters your body crosses a membrane using one of these mechanisms.



Membrane Transport: The Big Picture

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A&P REVIEW

Passive (No ATP)

Simple Diffusion

Small/nonpolar molecules move down gradient (O_2 , CO_2)

Facilitated Diffusion

Carrier or channel protein assists; still down gradient (glucose via GLUT)

Osmosis

Water moves toward higher solute concentration

Active (Requires ATP)

Primary Active

Pump uses ATP directly (Na^+/K^+ ATPase, H^+/K^+ ATPase)

Secondary Active

Uses ion gradient created by a pump (co-transport, antiport)

Vesicular

Endocytosis (into cell) and exocytosis (out of cell)

ACTIVITY

Instructions

With your table group, sort each scenario into the correct transport type. You have 5 minutes.

**Categories: Simple Diffusion | Facilitated Diffusion |
Active Transport | Osmosis | Endocytosis**

Be ready to explain WHY for each answer.

Sort These Scenarios

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ACTIVITY

A Oxygen moving from alveoli into blood

B Glucose entering a cell via a carrier protein, moving down its gradient

C Na⁺/K⁺ ATPase pumping ions against their gradient

D Water moving toward a region of higher solute concentration

E A white blood cell engulfing a bacterium

Choose:

- Simple Diffusion
- Facilitated Diffusion
- Active Transport
- Osmosis
- Endocytosis

Transport Sorting: Answers

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A O₂: alveoli → blood

Simple Diffusion

B Glucose via carrier, down gradient

Facilitated Diffusion

C Na⁺/K⁺ ATPase, against gradient

Primary Active Transport

D Water → higher solute

Osmosis

E WBC engulfs bacterium

Endocytosis (Phagocytosis)

**Every nutrient that enters your
body
crosses a membrane.**

The molecular machinery of transport is the foundation of digestion.

Stomach Anatomy: Regions

BIOL 40C

CONCEPT

Cardia

Surrounds the LES; entry point from esophagus

Fundus

Superior dome; often contains trapped gas

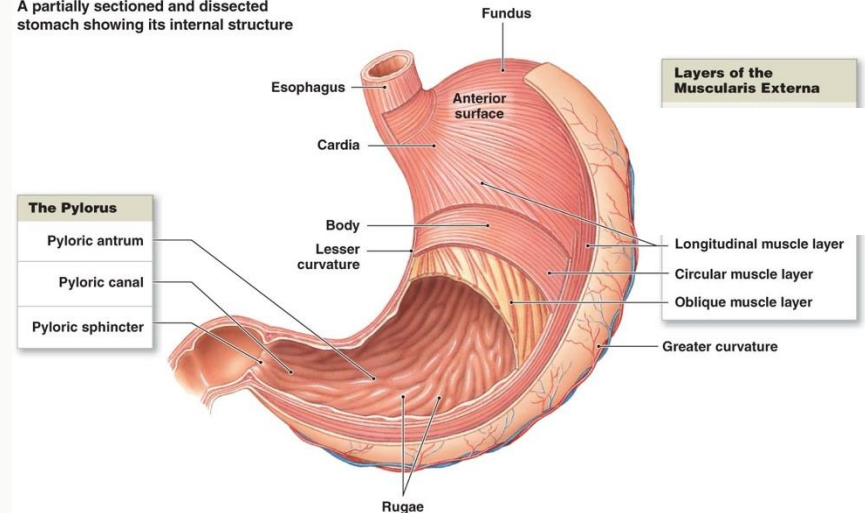
Body

Largest region; main site of churning and secretion

Pylorus

Funnel-shaped exit; contains pyloric sphincter

A partially sectioned and dissected stomach showing its internal structure



Stomach: Three Muscle Layers (Unique!)

BIOL 40C

CONCEPT

*Most of the alimentary canal has two muscle layers.
The stomach has three.*

1

Longitudinal (outer)

Runs along the length of the stomach

2

Circular (middle)

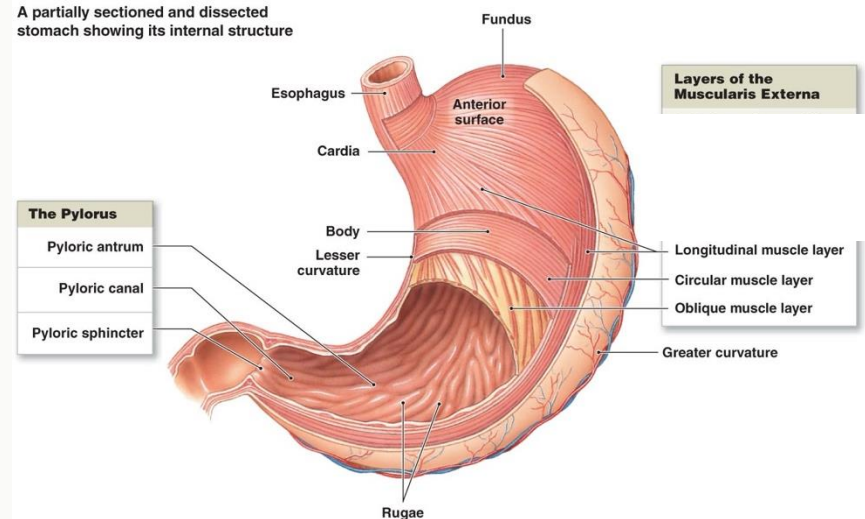
Wraps around the stomach circumference

3

Oblique (inner)

Unique to stomach; allows complex churning

A partially sectioned and dissected stomach showing its internal structure



Compare: Esophagus vs. Stomach

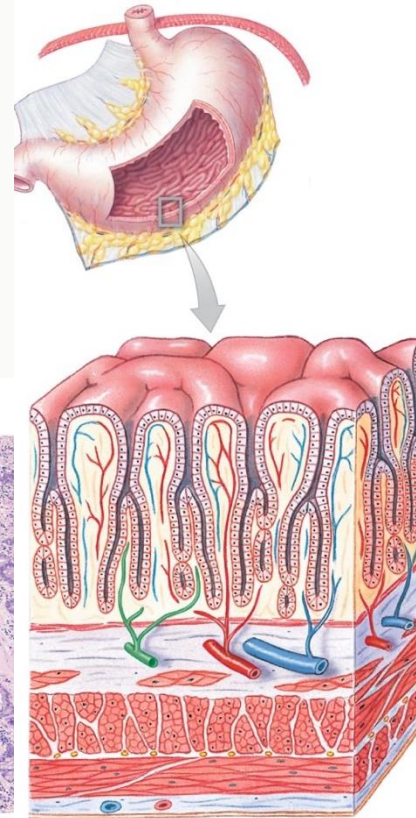
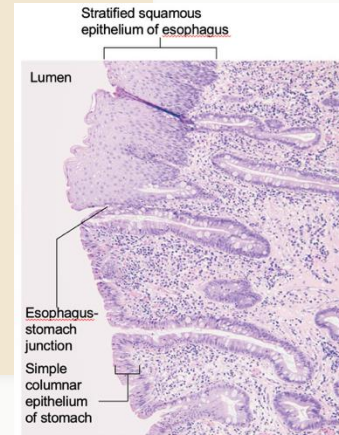
BIOL 40C

CHECK

**How does the structure of the stomach differ from the esophagus?
What does that tell you about the stomach's functions?**

Think about:

- Epithelium type (stratified squamous vs. simple columnar)
- Number of muscle layers (2 vs. 3)
- Primary function (transport vs. mixing + secretion)



BREAK

10 minutes
We'll resume at 11:05

Stomach Mucosa: Specialized Cell Types

BIOL 40C

CONCEPT

Simple columnar epithelium with deep gastric pits and glands

Mucous Neck Cells

Secrete thick protective mucus

Parietal Cells

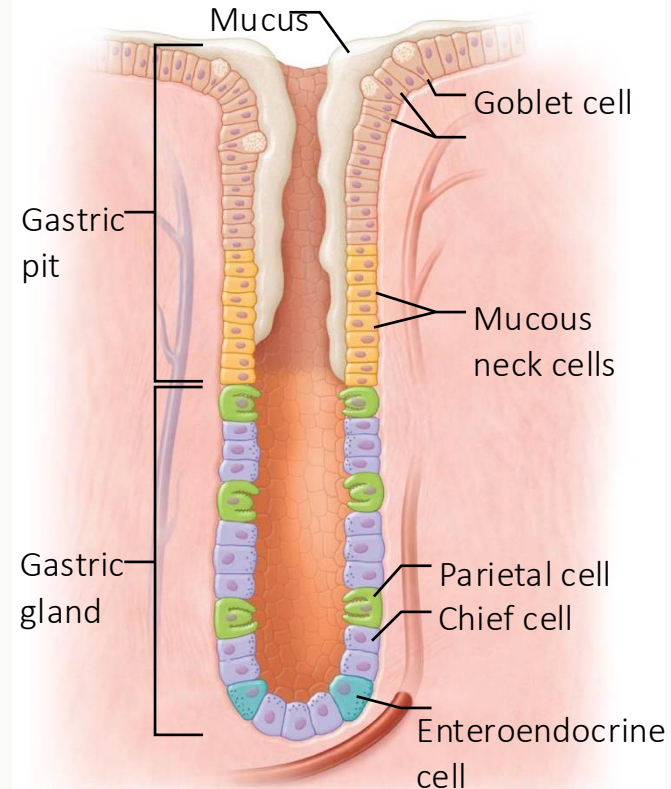
Secrete HCl (acid) + intrinsic factor (for B12)

Chief Cells

Secrete pepsinogen (inactive enzyme precursor)

Enteroendocrine Cells

Secrete hormones: gastrin, histamine, somatostatin



Stomach Mucosa: Specialized Cell Types

BIOL 40C

CONCEPT

Simple columnar epithelium with deep gastric pits and glands

Mucous Neck Cells

Secrete thick protective mucus

Parietal Cells

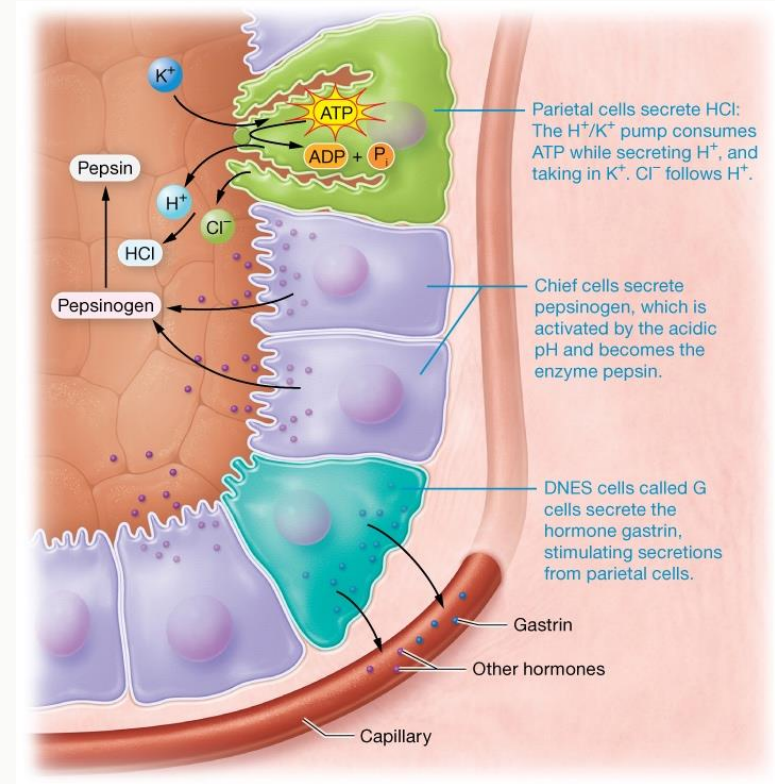
Secrete HCl (acid) + intrinsic factor (for B12)

Chief Cells

Secrete pepsinogen (inactive enzyme precursor)

Enteroendocrine Cells

Secrete hormones: gastrin, histamine, somatostatin



A&P REVIEW

This is where 40A transport concepts come alive.

The parietal cell uses multiple transport mechanisms you learned in 40A to produce stomach acid.

**We don't just say "the stomach makes acid."
We explain HOW, at the molecular level.**

This is the A&P perspective.

Parietal Cell: Step by Step (1/2)

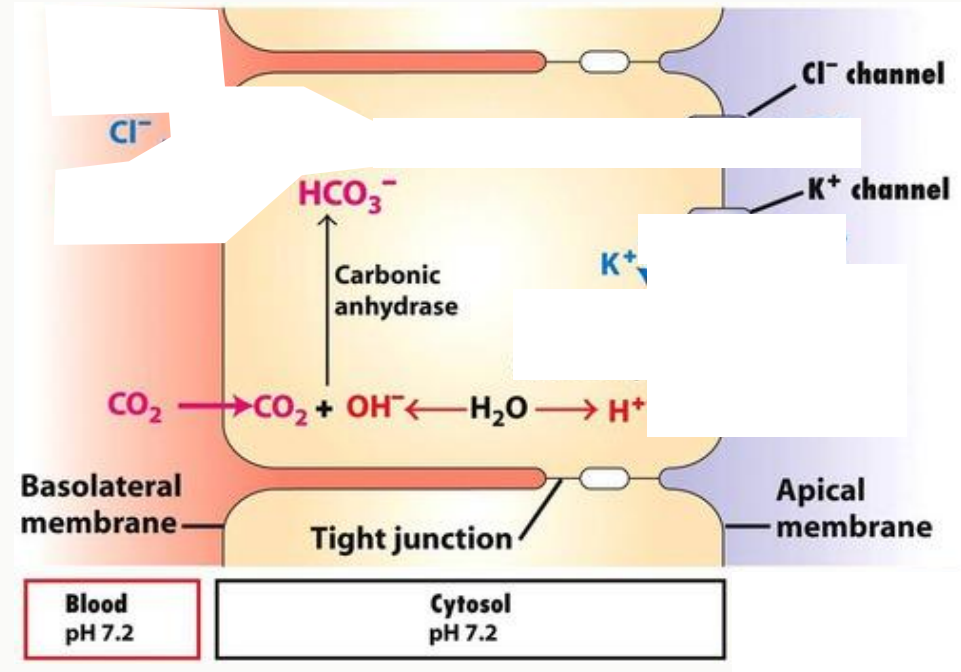
BIOL 40C

CONCEPT

1 CO_2 enters the cell by **simple diffusion**.
 $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$ (carbonic anhydrase) $\rightarrow \text{H}^+ + \text{HCO}_3^-$

2 H^+ pumped into lumen by H^+/K^+ ATPase (**primary active transport**, uses ATP)

3 K^+ recycled back out of the cell through K^+ leak channels (**facilitated diffusion**)



Parietal Cell: Step by Step (2/2)

BIOL 40C

CONCEPT

4

HCO_3^- exchanged for Cl^- via $\text{Cl}^-/\text{HCO}_3^-$ antiporter on the basolateral (**secondary active transport**)

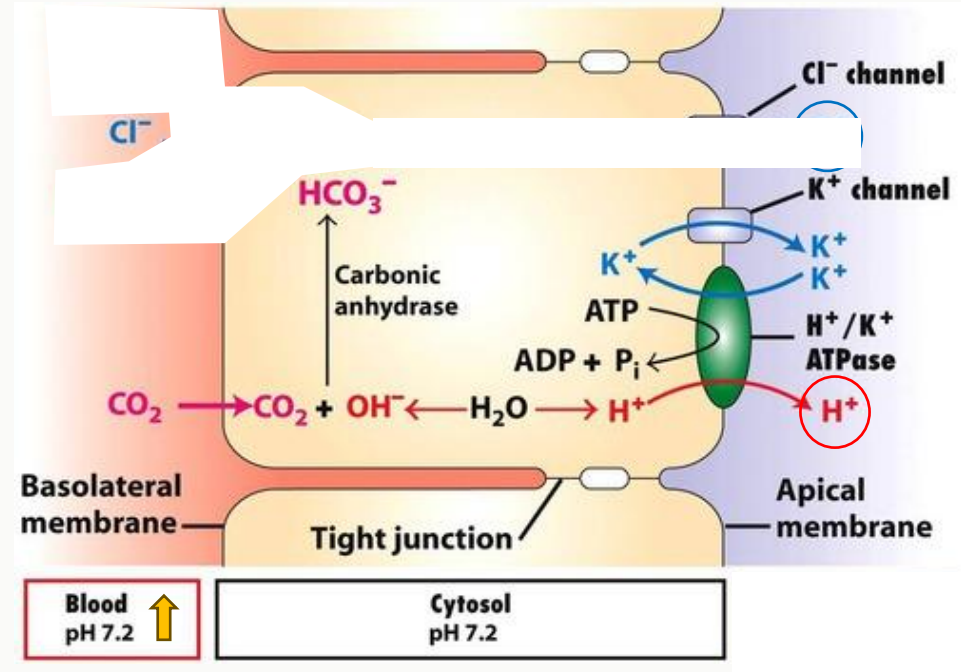
5

Cl^- moves through channels into the lumen (down electrochemical gradient; **facilitated diffusion**)

6

H^+ and Cl^- meet in the lumen $\rightarrow$ HCl forms OUTSIDE the cell (protects the cell!)

Alkaline Tide HCO_3^- entering the blood makes blood around the stomach slightly basic during active digestion.



CHECK

Which chemicals are moved by ACTIVE transport by parietal cells? (Select all that apply)

A H^+ (hydrogen ions)

B K^+ (potassium ions)

C HCO_3^- (bicarbonate)

D Cl^- (chloride)

Which chemicals are moved by ACTIVE transport by parietal cells? (Select all that apply)

0

H⁺ (hydrogen ions)

0%

K⁺ (potassium ions)

0%

HCO₃⁻ (bicarbonate)

0%

Cl⁻ (chloride)

0%

Active Transport: Answers

✓ H^+ — YES (H^+/K^+ ATPase = primary active)

✓ K^+ — YES (pumped by ATPase, recycled via channels)

✓ HCO_3^- — YES ($\text{Cl}^-/\text{HCO}_3^-$ antiport = secondary active)

≈ Cl^- — It depends! Antiport on basal side (active), but channels on apical (passive)

Where Exactly Does HCl Form?

CHECK

Choose the correct answer:

- ☐ A Inside the parietal cell cytoplasm
- ☐ B In the endoplasmic reticulum of parietal cells
- ☐ C Outside parietal cells, in the stomach lumen
- ☐ D In the mucous layer coating the stomach wall

Where exactly does HCl form?



Inside the parietal cell cytoplasm

0%

In the endoplasmic reticulum of parietal cells

0%

Outside parietal cells, in the stomach lumen

0%

In the mucous layer coating the stomach wall

0%

✓ C. Outside parietal cells, in the stomach lumen

H^+ is pumped out by the H^+/K^+ ATPase.

Cl^- exits through apical channels.

They combine in the lumen to form HCl.

This protects the parietal cell from its own acid!

The A&P Perspective

We don't just say "the stomach makes acid." We explain HOW, at the molecular level, using transport proteins you learned about in 40A. Structure dictates function.

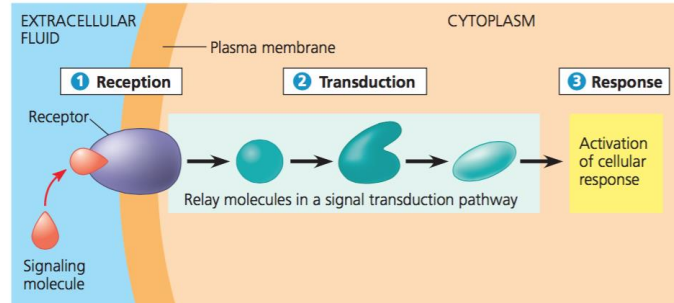
A&P Review: Signal Transduction

BIOL 40C

A&P REVIEW

The stomach doesn't just dump acid all the time. It's precisely regulated by signals.

Ligand-Receptor Signaling (from 40A)



A&P Review: Signal Transduction

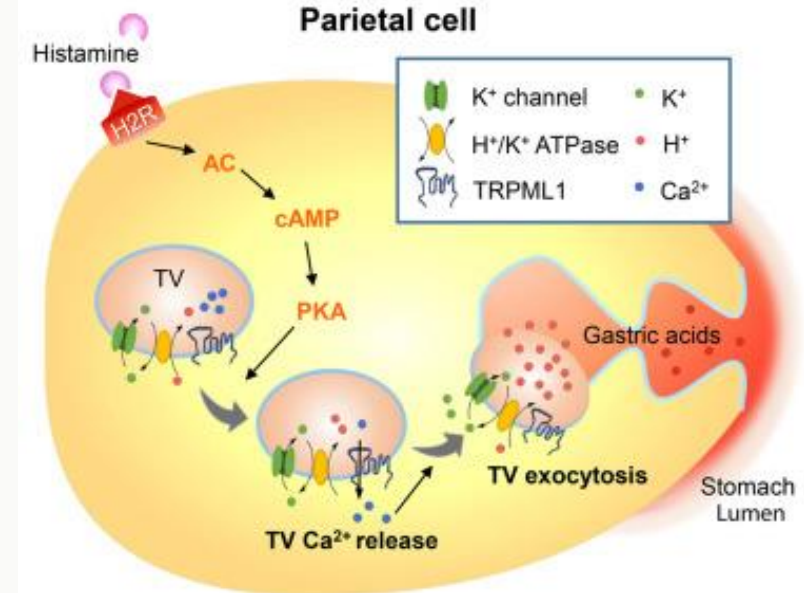
BIOL 40C

A&P REVIEW

*The stomach doesn't just dump acid all the time.
It's precisely regulated by signals.*

Example in the Stomach:

Histamine binds H₂ receptor on parietal cell → second messenger cascade → H⁺/K⁺ ATPase and K⁺ channels delivered to the membrane by **exocytosis** → acid secretion increases



Three Phases of Gastric Acid Secretion

BIOL 40C

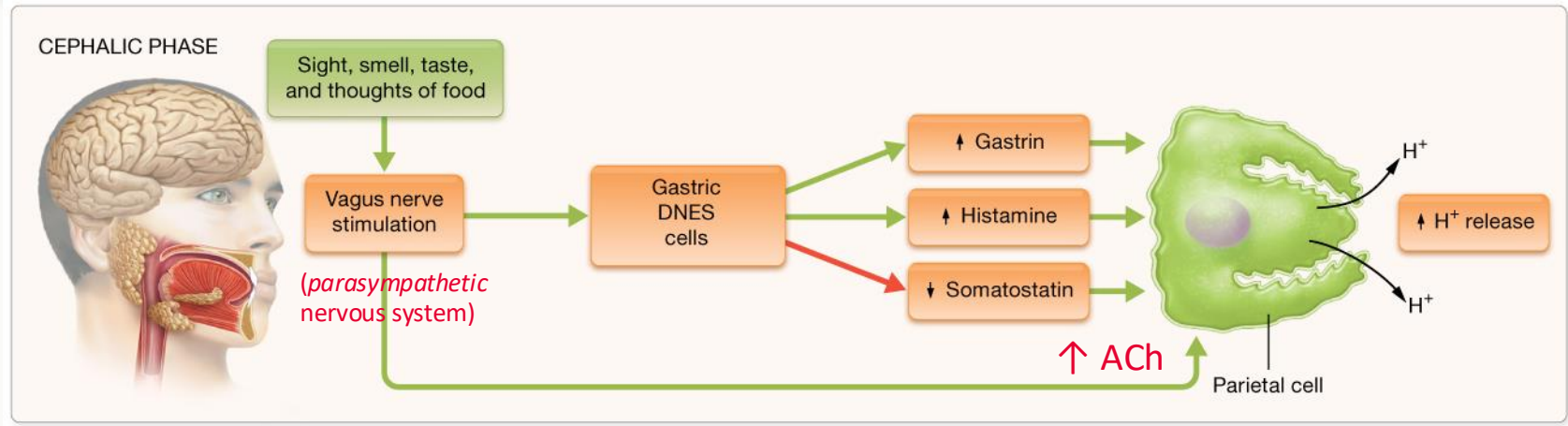
CONCEPT

Cephalic Phase

1

Trigger: Sight, smell, taste of food

Vagus nerve → ACh → stimulates parietal cells, gastrin, histamine



Three Phases of Gastric Acid Secretion

BIOL 40C

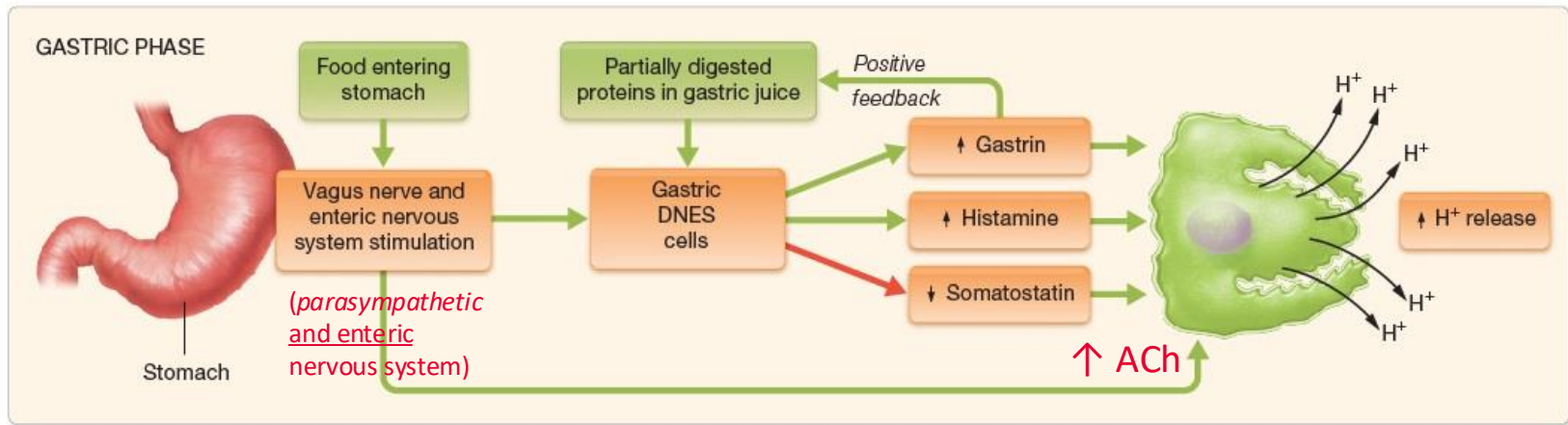
CONCEPT

Gastric Phase

2

Trigger: Food in stomach

Distension + protein fragments → ENS + gastrin → more acid



Three Phases of Gastric Acid Secretion

BIOL 40C

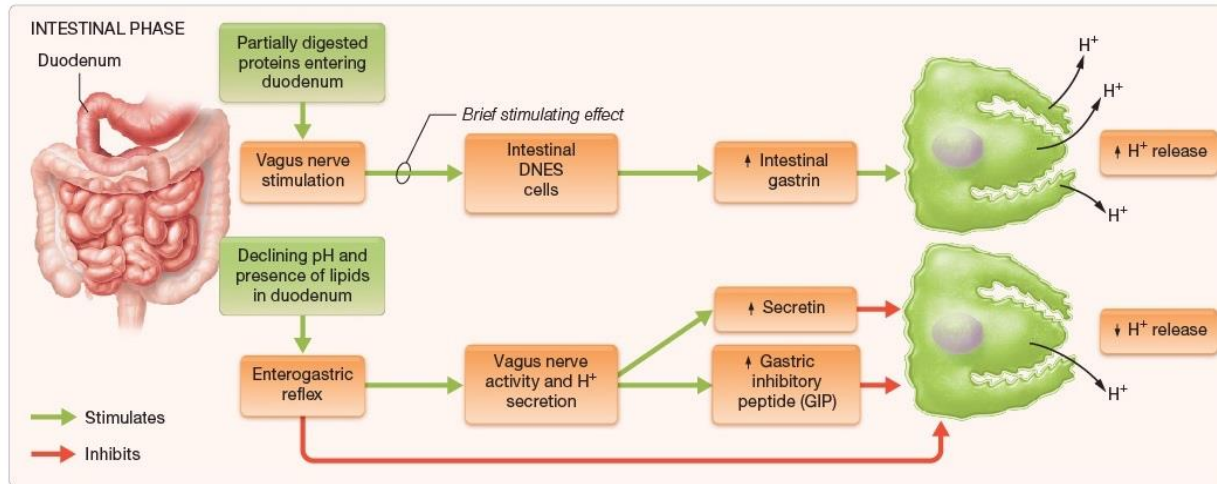
CONCEPT

Intestinal Phase

3

Trigger: Chyme enters duodenum

Brief stimulation, then inhibition (secretin, GIP shut it down)



CHECK

Which of the following would INCREASE acid production?

- A** Increased gastrin
- B** Decreased acetylcholine (ACh)
- C** Increased histamine
- D** Increased somatostatin

Which of the following would INCREASE acid production?

0

Increased gastrin

0%

Decreased acetylcholine (ACh)

0%

Increased histamine

0%

Increased somatostatin

0%

Answer: A and C

✓ Increased gastrin → stimulates parietal cells directly

✗ Decreased ACh → would DECREASE acid (ACh stimulates secretion)

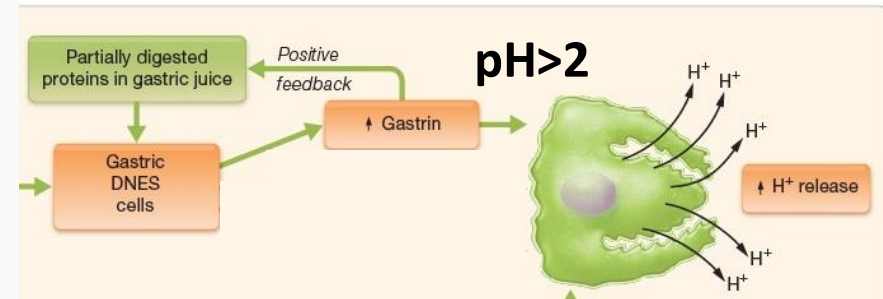
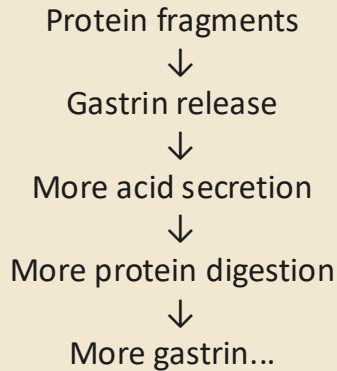
✓ Increased histamine → binds H₂ receptors, activates H⁺/K⁺ ATPase

✗ Increased somatostatin → would DECREASE acid (inhibitory hormone)

A&P REVIEW

Positive Feedback

(during active digestion)

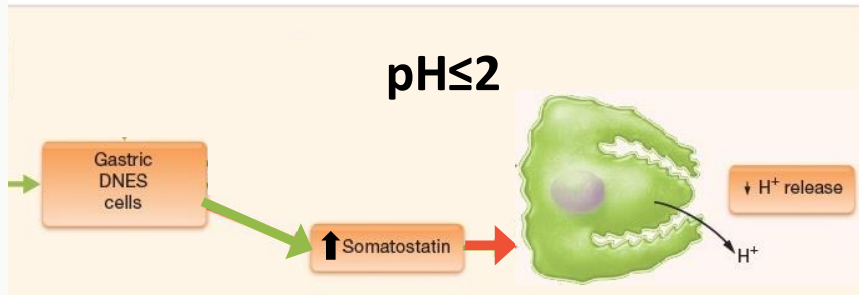


Master feedback loops now with the stomach. You'll see them in urinary, endocrine, and reproductive systems.

Feedback Loops in Acid Regulation

BIOL 40C

A&P REVIEW



Negative Feedback

(when $pH \leq 2$)

pH drops to 2 or below



Somatostatin released



Shuts down gastrin



Acid secretion stops



System resets

Master feedback loops now with the stomach. You'll see them in urinary, endocrine, and reproductive systems.

CONCEPT

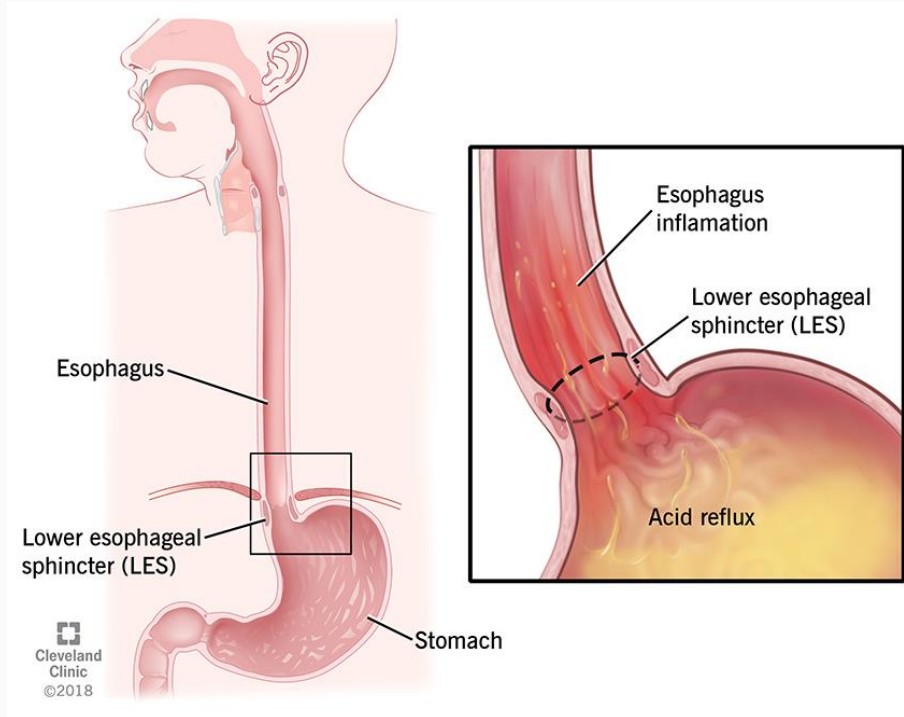
Gastroesophageal Reflux Disease

When the LES fails, acid contacts the esophageal mucosa. The stratified squamous epithelium is NOT designed to handle acid.

Result: pain, inflammation, erosion, and potentially Barrett's esophagus or cancer.

Risk Factors

Obesity, caffeine, alcohol, smoking, certain medications (NSAIDs), H. pylori infection, hiatal hernia



ACTIVITY

You're a pharmacologist.

You know how parietal cells work.
Design THREE different types of heartburn drugs,
each targeting a different point in the acid-production pathway.

*With your group, brainstorm for 2 minutes.
Think about what you just learned about the parietal cell.*

Hint: Where could you intervene in the molecular pathway?

Three Drug Strategies

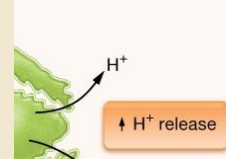
BIOL 40C

Antacids (e.g., Tums)

1

Neutralize acid already in stomach

After production



H2 Blockers (e.g., Zantac)

2

Block histamine receptor on parietal cell

At the signal level

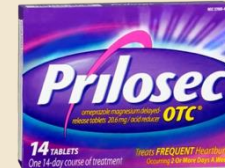
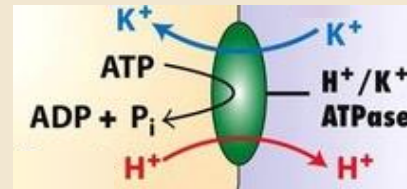


PPIs (e.g., Prilosec)

3

Block H^+/K^+ ATPase directly

At the pump itself



**Each drug targets a different level
of the molecular mechanism.**

This is the power of understanding A&P at the molecular level.

Before You Go: Weekend Assignments

BIOL 40C

ALW Workbook + Key Concepts Quiz

Due: Sunday, April 12 at 11:59 PM

2 attempts on the quiz, highest score kept.

Use the workbook to study, then take the quiz.

Coming Monday (April 13):

We'll debrief your KC Quiz results and dive into the small intestine.

The more prepared you come, the more you'll get from class.

Questions? Office hours are listed on Canvas.